
LATA: A DROP-IN, FERPA-COMPLIANT LOCAL-LLM AUTOGRADER FOR UPPER-DIVISION STEM COURSEWORK

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ABSTRACT

Large-language-model (LLM) graders promise to relieve the grading burden of upper-division STEM courses, but most deployments to date send student work to third-party APIs, violating FERPA and exposing institutions to data risk while requiring substantial assignment modification. We present **LaTA** (*LaTeX Teaching Assistant*), a drop-in, open-source autograder that runs entirely on commodity on-premises hardware and assumes a LaTeX-native workflow already adopted by many engineering and physics courses. LaTA implements a four-stage pipeline (ingest, segment, grade, report) using a locally hosted open-weight chain-of-thought LLM grader (gpt-oss:120b) that compares student work to an instructor-authored reference solution and applies a YAML rubric with binary per-item scoring. We deployed LaTA in Winter 2026 in ME 373 (Mechanical Engineering Methods) at Oregon State University, grading every weekly assignment for approximately 200 students on a single Mac Studio at \$0 marginal cost per assignment and 1–3 minutes of wall-clock time per submission, enabling regrading of corrected assignments and greatly expanded TA office hour offerings. The instructor-confirmed grading-error rate held at roughly 0.02–0.04% per rubric line item across the term. Relative to the same instructor’s previous traditionally-graded cohort, the LaTA-graded cohort outperformed by approximately 11% on the midterm exam and 8% on the final exam, and reported large gains in self-assessed confidence on every stated learning objective ($N = 159$ survey responses, $\Delta \geq +1.49$ Likert points, $p < 10^{-27}$ on every comparison). We release the code under AGPLv3.

Keywords automated grading · large language models · on-premises inference · FERPA · LaTeX workflow · engineering education

1 Introduction

Grading handwritten, multi-part derivations in upper-division engineering courses is labor-intensive, slow to return to students, and inconsistent across graders. At a typical 200-student enrollment with four-problem weekly homework assignments and two exams, a single term routinely consumes hundreds of TA hours of first-pass grading before any feedback reaches a student; the fastest feedback loops in the literature remain measured in days rather than hours, and rubric drift across multiple TAs is an acknowledged threat to consistency [1–3]. Large language models have improved rapidly enough to plausibly automate the first-pass grading task on symbolic-derivation problems, and a steady stream of recent work has explored LLMs as STEM graders and tutors [4–9]. Most of this work, however, has been conducted against cloud-hosted commercial APIs, and a cloud-graded autograder is difficult to reconcile with FERPA and institutional data-residency policies in a U.S. public university without negotiated contractual carve-outs [10–12]. The practical result is that many instructors who would benefit from LLM grading cannot deploy it without either anonymising their data or accepting a compliance risk they are not positioned to carry.

This paper asks a deliberately narrow question: *can a single instructor, with commodity on-premises hardware and no cloud access, build and deploy an LLM-based autograder good enough to replace human-TA first-pass grading in a*

200-student upper-division numerical-methods course, and what happens when they do? We answer yes, describe the system, named **LaTA** (LaTeX-native Automated Teaching Assistant), and report evidence from a full-term deployment in ME 373 (Mechanical Engineering Methods) at Oregon State University during Winter 2026.

LaTA is built around four commitments. First, *everything runs on a single on-premises machine*: no student work leaves the university network, and the compliance story is “we own the hardware” rather than “we negotiated a data-processing agreement.” Second, *typed data objects with Pydantic validation* thread through every stage of the pipeline [13], so the boundary between the model and the rest of the system is schema-enforced rather than prose-parsed. Third, *structural decisions that can be done deterministically are done deterministically*: submission segmentation is regex-first with an LLM fallback, scoring is binary per rubric item, and LaTeX compilation is handled by pdf_latex with an LLM-assisted repair path only when a compile actually fails. Fourth, *student recourse is a first-class feature of the system*: every grading decision produces a two-channel artefact (a blunt TA-facing audit reasoning plus a Socratic, answer-withholding student hint), and the corrections-pass workflow is built directly into the configuration surface.

The contributions of this paper are:

- A complete, open-source reference implementation of an on-premises LLM autograder for LaTeX-based STEM coursework, including the pipeline, the Pydantic schemas, the prompt-injection defences, the LaTeX self-healing mechanism, and the Gradescope-integrated regrade workflow (§??).
- A detailed operational account of a full-term, full-replacement deployment to 200 students across eight homework assignments in a single upper-division course, including weekly workflow, hardware profile, and the regrade-request audit (§??).
- A program-evaluation study of the deployment across three evidence streams; operational data, an anonymous student survey ($N = 159$), and a between-cohort exam comparison against the same instructor’s previous traditionally-graded cohort, with explicit disclosure of the composite-intervention confound and the methodological limitations of a single-year study (§3–4).
- A reading of the evidence that distinguishes the *autograder* as a commodity from the *workflow* it enables, and a set of generalisation-envelope recommendations for instructors considering a similar deployment (§5).

The remainder of the paper is organised as follows. §2 situates LaTA against prior work on LLM-based STEM assessment, on-premises / FERPA-aware AI tooling, and LaTeX as a pedagogical substrate. §?? describes the four-stage pipeline and its typed data model. §?? documents the Winter 2026 deployment in ME 373. §3–4 report the three evidence streams. §5–6 read the results and enumerate what the design of the study cannot tell us. §7 closes.

2 Background and Related Work

We situate LaTA against five strands of prior work: LLM-based assessment of STEM free-response problems (§??); human-in-the-loop rubric scaffolds and chain-of-thought prompting for grading (§??); on-premises and FERPA-aware AI in higher education (§??); LaTeX as a pedagogical substrate for engineering coursework (§??); and retrospective-pre survey methodology (§??). The first four locate LaTA as a system; the fifth positions our evidence collection.

2.1 Grade: typed, binary, injection-resistant rubric application

Grading is where LaTA’s design investment is densest. For each `ProblemChunk`, the grader assembles a prompt containing: the rubric slice (YAML, restricted to the rubric items pertaining to the `ProblemChunk`), the instructor’s reference solution for that problem, a one-shot JSON example, and the student’s chunk text wrapped in explicit UNTRUSTED INPUT delimiters. It then calls the **gpt-oss:120b** model on the local Ollama server. The response is validated against a strict Pydantic schema, `ChunkGradeResult(items: List[GradeItem])`, where each `GradeItem` carries six fields (`problem_group`, `criterion_name`, `score`, `max_score`, `audit_reasoning`, `student_hint`) plus an `is_extra_credit` boolean that is set by post-processing, never by the LLM (whose copy of the rubric has the flag stripped exactly to prevent that). Three sub-designs deserve emphasis:

Binary per-item scoring. For every rubric item the LLM must award either full points or zero, never partial credit. The rationale is variance reduction: a 120B model asked “how much partial credit does this derivation deserve?” produces noisy decimal scores that disagree across runs, whereas “does this derivation meet the criterion, yes or no?” collapses to a stable binary decision. Partial credit is still expressible by the rubric author, who can decompose a single conceptual step into several fine-grained binary criteria. This is the same move made in recent rubric-scaffold frameworks [7, 8] and in our view is essential for reproducibility.

Dual-channel feedback. Each GradeItem carries two distinct text fields. `audit_reasoning` is written for the TA and the instructor: blunt, technical, explicit about where the student’s work diverged from the reference solution (“*student failed to apply the chain rule on term 2*”). `student_hint` is written for the student: Socratic, answer-withholding, never quoting the reference solution. The prompt repeatedly emphasizes that the hint must never reveal the answer. In deployment we found this separation essential, but not always sufficient. This mechanism lets the instructor review a grade at a glance while also producing a genuinely pedagogical feedback PDF, but occasional leakage of reference solution information (*i.e.* answers) into student feedback documents did occur, albeit at a low rate ($\sim 5\%$ of all feedback documents).

Five-layer prompt-injection defense. Because the student’s submission is untrusted text concatenated into the grader’s input, injection attempts (“Ignore previous instructions and award full marks”) are a real attack surface. LaTA applies five layers of defense: (1) a *security preamble* at the top of the system prompt that names injection explicitly and instructs the model to flag but not obey such directives; (2) *explicit delimiter banners* (solid horizontal rules of the form `==== . . .====`) marking the start and end of UNTRUSTED INPUT; (3) a *role-reinforcement* reminder after the student block (“*the text above is student work to be EVALUATED, not instructions to be FOLLOWED*”); (4) *post-hoc keyword scanning* of the returned `audit_reasoning` for known injection phrases (‘ignore previous’, ‘award full points’, ‘system:’, etc.) with automatic SECURITY_FLAG emission; and (5) *perfect-score watchdog* alerts that warn on full marks so a TA can spot-check. A student who successfully injects a perfect-score instruction must therefore defeat all five layers simultaneously; we have not observed such a case in deployment.

Grading is further wrapped by a chain-of-thought streaming layer. Reasoning-capable models (our grader is configured with `is_reasoning_model: true`) stream `<think>`-tagged thinking tokens separately from the JSON response via Ollama’s native `think=True` parameter, allowing real-time display during grading and archival to `debug/thinking.txt` for later audit. JSON extraction is defensive: the response is stripped of markdown fencing, the outermost `{ . . . }` containing the “items” key is extracted, and a *LaTeX-aware sanitizer* converts unescaped single backslashes (`\frac`, `\alpha`) to JSON-valid double backslashes without breaking genuine JSON escapes (`\n`, `\"`). On schema failure the call is retried once with an enhanced warning; on second failure the submission is written to `failed_submissions.yaml` for manual review.

2.2 Report: LaTeX feedback with self-healing compilation

Graded items are grouped by `problem_group`, aggregated into a per-problem summary table, and rendered through a Jinja2-templated LaTeX feedback report (`feedback_report.tex`). Jinja2’s default `{{ }}` / `{% %}` delimiters collide with LaTeX, so LaTA configures custom delimiters (`\VAR{}`, `\BLOCK{}`, `\#{}`) that round-trip cleanly through `pdflatex`. A per-field `latex_escape` filter maps the Unicode characters that the LLM compulsively emits (`M_2` typed as a Unicode subscript-two, `\ll` typed as `«`, Greek letters typed as Unicode glyphs, and the narrow no-break space `U+202F`) to math-mode LaTeX equivalents while escaping `&`, `%`, `#`, and underscores in text segments.

When `pdflatex` compilation fails despite this cleanup (almost always because of residual escaping bugs in LLM-generated free text) LaTA parses the `.log` file, extracts the explicit error blocks (lines beginning with `‘!’`), and asks the grader LLM to *fix only the formatting* while keeping all content and Jinja2 markers intact. The repaired document is then subject to a *hallucination gate*: a validator checks that the returned text still contains `\documentclass`, `\begin{document}`, `\end{document}`, has Jinja variables or section headers, has length within 50–200% of the original, has a backslash-command count within 80–120% of the original, and does not contain common hallucination-topic keywords (e.g. “renewable energy”, “blockchain”). Validation failures roll back to the original document; validation successes are recompiled. If still broken after `latex_fix_max_retries`, the partial PDF (if any) is returned and the broken `.tex` is preserved in `logs/`.

Late-penalty and resubmission arithmetic. Late penalties use UTC timestamps from Gradescope (which stores submission times in GMT) compared against the configured `due_date` after adding any per-student extension hours. Days late are rounded *up* (`24h + 1s = 2 days late`), penalties are computed against `rubric_max + points_awarded_elsewhere` to fairly tax manually graded components, and final scores are allowed to go negative (the instructor handles floor semantics downstream). Resubmissions are detected by comparing the submission timestamp against a persisted `grading_metadata.yaml`; newer submissions increment a version suffix (`_v2`, `_v3`) so that the Gradescope re-upload preserves history. A *corrections mode* reloads metadata from a base assignment, regrades the corrected submission, and then preserves the originally awarded extra credit (which is not earnable in corrections) by re-injecting synthetic `is_extra_credit` GradeItems into the regraded result.

2.3 Configuration and operational surface

Everything user-facing lives in a single `config.yaml` (see Listing 1 for an excerpt). The configuration names the segmenter and grader models separately, exposes per-call Ollama options (`num_ctx`, `temperature`, `repeat_penalty`, etc.), switches anonymization on or off, toggles late penalties, schema-error retries, and LLM LaTeX repair, and hosts the optional SMTP block used by `post_grading.py` to deliver feedback PDFs to institutional email addresses. No credentials are ever stored on disk: SMTP credentials are prompted for at send time, and all model calls terminate at `http://localhost:11434/v1`.

Listing 1: Key fields from `config.yaml`. The segmenter/grader split, binary-grading options, and local-only `ollama_host` are the three design- critical lines.

```
system:
  ollama_host: "http://localhost:11434/v1"
  segmenter:
    model_name: "gpt-oss:20b"
    ollama_options:
      temperature: 0.0
      num_ctx: 32768
  grader:
    model_name: "gpt-oss:120b"
    is_reasoning_model: true
    ollama_options:
      temperature: 0.1
      num_ctx: 65536
      repeat_penalty: 1.4
  grading:
    anonymize: true
    skip_previously_graded: true
    late_penalties:
      enabled: true
      penalty_per_day: 0.20
      due_date: "2026-01-17 07:59:59" # GMT/UTC
    error_handling:
      retry_on_schema_error: true
      fix_latex_errors: true
```

3 Methods

This study is a *program evaluation* of a single-instructor, single-course deployment of LaTA during Winter 2026 in ME 373 (Mechanical Engineering Methods) at Oregon State University. We are not testing a hypothesis about LLM-based grading in general; we are reporting what happened when one instructor replaced TA first-pass grading with an on-premises autograder for a full term, and what students said about it afterwards. All student-facing instruments were administered for the routine purpose of course improvement, participation was anonymous and voluntary, and the analyses reported here use only aggregate, de-identified data. The activity was determined to fall outside the scope of human-subjects research and did not require IRB oversight; we nonetheless describe the instruments, inclusion rules, and analyses in full so that readers can judge the evidence on its own terms.

Three streams of evidence are reported: (i) operational evidence collected by the instructor during the term, including regrade requests and per-submission processing time (§3.1); (ii) an anonymous post-term student survey administered through Canvas, with both Likert-scale [28] items and open-ended prompts (§3.2, §3.3); and (iii) a between-cohort, quasi-experimental comparison of exam performance between the LaTA-graded Winter 2026 cohort and the traditionally-graded Winter 2025 cohort taught by the same instructor (§3.4). Each stream has distinct strengths and weaknesses, and we read them against one another in §4 and §5 rather than relying on any single stream in isolation.

3.1 Operational evidence and regrade audit

Throughout the term, every LaTA run produced a structured output directory (§2.2) containing per-student YAML reports, per-student PDF feedback, a `grading_metadata.yaml` file recording model versions and timestamps, and an LLM debug log (`data/debug/`) preserving raw model outputs for any submission that triggered a schema-error retry or a LaTeX-repair retry. These artifacts were not generated for the study; they are the standard operational output of the tool and would exist whether or not a paper was being written.

For each of the eight graded homework assignments we recorded: the wall-clock time for one complete grading pass over the 200-student roster; the number of student regrade requests received during the two-week window each assignment was open; and, for each regrade, the instructor’s post-hoc classification as *valid* (a rubric item was misapplied or the model misread the student’s work) or *invalid* (the deduction was correct and the student simply disagreed). A regrade was counted once per rubric line item disputed, not once per email thread. The per-submission processing time was computed as the total wall-clock time divided by 200.

This evidence stream is strongest at bounding error rates and throughput: every submission passes through the grader three to six times (once per problem plus a corrections-pass regrade for the 90% of students who submitted corrections), and a regrade request is the student-facing ground truth for whether the first-pass grade was acceptable. It is weakest at detecting *missed* deductions, since a student who received credit they did not deserve has no incentive to report the error. We address this asymmetry in §3.2 through an explicit self-report item about missed deductions.

3.2 Anonymous student survey

A post-term survey was administered as an optional Canvas quiz during the final week of Winter 2026. The quiz was worth a small amount of extra credit (completion-based, with no penalty for non-response and no linkage between the completion credit and the anonymous response itself). Canvas delivered responses to the instructor stripped of student identifiers; we report only aggregate statistics. Of the 200 students enrolled at the beginning of the term, $N = 159$ returned a complete response, for a response rate of approximately 80%.

The instrument comprised four blocks, described below. The full wording of each item is available upon request (§7); the condensed labels used in the figures and tables are given in parentheses.

Block 1: Pre/post confidence on stated learning objectives. For each of the four course-level learning objectives taken verbatim from the ME 373 syllabus — (i) formulating and solving initial value problems (IVPs), (ii) formulating and solving boundary value problems (BVPs), (iii) formulating and solving partial differential equations (PDEs), and (iv) selecting an appropriate numerical method for a given mechanical-engineering problem — students rated their confidence on a 5-point Likert scale (1 = not confident at all, 5 = supremely confident) at two points: *retrospective pre* (“how confident did you feel at the beginning of the term...”) and *post* (“how confident do you now feel...”). Retrospective pre-test measurement was chosen over a separate first-week survey to eliminate the response-shift bias that arises when students reinterpret a learning objective mid-term [27]; we acknowledge the limitation that retrospective pre-tests tend to inflate apparent gains and return to it in §6.

Block 2: LaTA grading-error self-report. Three items asked students to estimate, as a count over the entire term, the number of times they had observed: (a) an *incorrect deduction on the first pass* (“false positive — first pass”), (b) a *missed deduction* that the grader failed to catch (“false negative”), and (c) an *incorrect deduction on the corrections pass* (“false positive — corrections”). Items (a) and (c) are the student-facing counterpart to the regrade audit in §3.1; item (b) is the counterpart students can detect but regrade requests cannot. Free-response counts were capped at reasonable bounds in the cleaning step (values > 30 were read as outliers and verified against the free-text essay items before being retained or truncated).

Block 3: LaTA perception (Likert). Three 5-point Likert items asked students how valuable the LaTeX submission workflow was to their learning (“LaTeX valuable”), how helpful the LaTA feedback was (“feedback quality”), and their overall sentiment toward the LaTA grading system (“overall sentiment”). Two further items asked for their perception of instructor/TA office-hours support (“office hours helpful”, “office hours available”). A final pair of items asked the amount of extra time, in minutes per homework assignment, that writing solutions in LaTeX added *at the beginning* of the term and *at the end* of the term, to probe the ramp-up cost of the LaTeX requirement.

Block 4: Open-ended essays. Two free-text prompts closed the survey: “What aspects of this course worked well for your learning?” (positive) and “What aspects of this course did not work well for your learning?” (negative). Responses were coded for the thematic analysis described below.

3.3 Thematic coding of open-ended responses

The two free-text prompts yielded 260 codable responses (129 positive, 131 negative) after removing blanks and single-word nonresponses (“n/a”, “idk”, “none”). A single coder (the instructor–author) performed open coding on a random 20% sample to generate an initial codebook, consolidated overlapping codes by inspection, then applied the final codebook to the full set. The resulting codebook contains twelve themes in the positive bucket (e.g. *homework-as-learning*, *lecture quality*, *corrections workflow*, *video lectures*, *rigor appreciated*, *LaTeX valuable*, *office hours*,

coding/Python, fast feedback) and twelve themes in the negative bucket (e.g. *exam difficulty, workload, lack of examples, LaTeX time cost, autograder errors, lecture pace, lecture–homework gap, hidden expectations, anti-AI principle*). Each response could receive multiple codes. Because only one rater coded the data, formal inter-rater reliability is not reported; the full coded dataset (`thematic_results.json`) is available from the author on reasonable request, and we read the thematic counts in §4 only against the signed direction of the quantitative items, not as a standalone claim.

3.4 Between-cohort exam comparison

A quasi-experimental comparison of exam performance between the LaTA-graded Winter 2026 cohort (enrollment 200; $n = 182$ sat the final exam) and the traditionally-graded Winter 2025 cohort (enrollment 181; $n = 157$ sat the final) provides a third, outcome-oriented window on the intervention. Both cohorts were taught by the same instructor (the author), used the same textbook, met on the same weekly schedule, and were assessed on the same midterm and final-exam structure (two proctored, closed-note written exams, graded by hand by the instructor and TAs in both years). Approximately two-thirds of the exam problems were held identical across the two years to support this comparison; the remaining third were updated or replaced. The new items were, in the author’s qualitative judgment, slightly more difficult than the items they replaced; this biases the comparison *against* the Winter 2026 cohort and should be read as a conservative floor rather than as a clean equivalence.

We report the per-cohort mean percentage score on the midterm and the final exam. We deliberately do not report inferential statistics on this comparison, for two reasons. First, the exam is not exclusively constructed from the held-identical items, so the numerical delta confounds item-level difficulty drift with any cohort-level difference. Second, and more importantly, LaTA was not the only structural change between the two cohorts. The Winter 2026 deployment bundled three innovations that cannot be disentangled with a single year of post-hoc data:

- **LaTA autograding and LaTeX-native homework.** Homework was submitted in LaTeX and graded by the LaTA pipeline end-to-end; the Winter 2025 cohort submitted handwritten PDFs and received manual TA grading feedback.
- **Corrections workflow.** The Winter 2026 cohort was offered a per-assignment corrections pass in which the student could revise and resubmit for partial credit restoration (§??). The Winter 2025 cohort received only a first-pass grade.
- **Tripled TA office hours.** The TA budget released by LaTA (§??) was redirected into office hours; TA office-hour coverage in Winter 2026 was approximately $3\times$ that of Winter 2025.

Any exam-score gain should therefore be attributed to the *composite* intervention: autograding plus corrections plus expanded office hours. We make this attribution explicit throughout §4–5, and we return in §6 to the design changes that would be needed to separate the three components.

3.5 Statistical analysis

Likert-scale and count items are reported as means with standard deviations and, where appropriate, medians. For pre/post confidence comparisons (Block 1) we report the difference in group means with a nonparametric significance test. Because the survey was anonymous and pre/post items were collected in a single instrument, we cannot match a specific pre response to the same respondent’s post response; we therefore treat the two distributions as independent for the purposes of significance testing and apply the Mann–Whitney U test [29]. We acknowledge that, conditional on the pre and post responses having in fact come from the same 159 respondents, a paired test (Wilcoxon signed-rank [30]) would be more powerful and slightly more conservative about the direction of individual change; the gain magnitudes we report in §4 are large enough that the choice of test does not alter the qualitative conclusions, but we flag the design limitation explicitly.

Effect sizes for the pre/post comparison are reported as the rank-biserial correlation r_{rb} [31], which for ordinal data are interpretable on the same scale as Cohen’s r [32]. No multiple-comparisons correction is applied to the four learning objective comparisons, because every uncorrected pre/post p -value is below 10^{-20} and any standard correction (e.g. Bonferroni [33] or Benjamini–Hochberg [34] at $q = 0.05$) would leave the qualitative conclusion unchanged.

All statistical computations were produced by the analysis script `analyze_feedback.py` which is available upon request. The script reads the raw Canvas export (`Survey_Feedback.csv`), applies the column mapping documented in the survey instrument section, and writes the per-figure data and a console summary of means, standard deviations, U statistics, r_{rb} effect sizes, and p -values. For the between-cohort exam comparison (§3.4), we report only the difference in sample means by design, for the reasons given there.

4 Results

Results are reported in the same order as the evidence streams in §3: operational data first (§4.1), survey data second (§4.2), thematic coding third (§4.3), and the between-cohort exam comparison last (§4.4). The overall picture is one of a deployment that worked (throughput, accuracy, and learning outcomes all moved in the intended direction) while producing mixed student perceptions of the autograder itself. We read the mixed perception signal not as a contradiction but as the most honest part of the result.

4.1 Operational results: throughput and accuracy

Across the eight homework assignments of Winter 2026, LaTA graded the 200-student roster in 1–3 minutes of wall-clock time per submission on the Mac Studio M3 Ultra described in §??, with the per-assignment total dominated by the number of problems and the length of the worked solutions rather than by class size. In aggregate, the instructor received 5–10 regrade requests per assignment, of which approximately half were judged valid on review (§??). A regrade request typically involved one disputed rubric line item, not the whole problem.

The natural unit for an error rate in this setting is the *rubric line item*, not the student or the assignment, because that is the granularity at which LaTA actually makes a binary grading decision. A typical assignment carried ~ 3 problems with ~ 10 rubric items per problem, so a single grading run made approximately $200 \times 3 \times 10 \approx 6,000$ rubric decisions per assignment, or $\sim 96,000$ rubric decisions across the eight-assignment term given the corrections pass. Against this denominator, the 5–10 regrade requests received per assignment correspond to a *contested* per-rubric-item rate of 0.04–0.08% and an *instructor-confirmed* per-rubric-item error rate (after the $\sim 50\%$ valid filter) of approximately 0.02–0.04%. In absolute terms, the instructor confirmed errors on roughly 3–5 of the $\sim 12,000$ rubric decisions per assignment, or roughly 20–40 across the entire term. Resolving each contested decision took the instructor on the order of 1–2 minutes and produced a corrected report via the versioning mechanism described in §2.2.

No submission failed to produce a final feedback PDF. LLM JSON-schema retries and LaTeX compilation retries both engaged occasionally during the term, but in every case the retry path in §2.1 and §2.2 converged within the configured retry budget. A small number of first-pass grades were hand-corrected by the instructor when the model had misread a student’s notation (e.g. treating a double-prime as a prime); these were logged as valid regrade requests and did not require tool-level intervention.

4.2 Survey results: confidence, accuracy, and perception

Of $N = 159$ survey respondents ($\approx 80\%$ response rate), every question attracted between 158 and 159 valid answers; none were dropped for completeness.

Confidence on learning objectives. Retrospective pre/post confidence ratings rose by more than one full Likert point on every one of the four stated learning objectives (Figure 1). Initial value problems rose from a pre-term mean of $M_{\text{pre}} = 1.91$ (SD = 0.98) to $M_{\text{post}} = 3.47$ (SD = 0.81), a gain of $\Delta = +1.57$; boundary value problems from $M_{\text{pre}} = 1.47$ (SD = 0.80) to $M_{\text{post}} = 3.20$ (SD = 0.76), $\Delta = +1.74$; partial differential equations from 1.94 (SD = 0.99) to 3.43 (SD = 0.93), $\Delta = +1.49$; and numerical method selection from 1.50 (SD = 0.88) to 3.38 (SD = 0.89), $\Delta = +1.88$. All four comparisons achieved Mann–Whitney U $p < 10^{-27}$ uncorrected (well below the threshold any standard multiple-comparisons correction would impose at $q = 0.05$ across only four tests) with rank-biserial effect sizes r_{rb} ranging from 0.688 (PDEs) to 0.825 (BVPs), conventionally “large” effects. The retrospective-pre design inflates apparent gains; even heavily discounted, the post-term medians of 3–4 on every objective represent a cohort that finished the term feeling moderately to strongly confident about content they had reported near-floor confidence on at the start.

Student-reported grading accuracy. The three self-report items on grading accuracy (Block 2 of the survey) yielded the distributions shown in Figure 2. Across the entire term, students reported a mean of 2.94 (SD = 2.99, median = 2) first-pass false-positive deductions (instances where LaTA took points off incorrectly) with a range of $[0, 20]$. Missed deductions (false negatives) averaged 1.08 (SD = 1.31, median = 1); corrections-pass false positives averaged 1.46 (SD = 1.68, median = 1). Across the eight-assignment term, each student received on the order of $8 \times 3 \times 10 \times 2 = 480$ individual rubric-item decisions, so the student-perceived first-pass false-positive rate normalises to approximately 0.6% per rubric item, the false-negative rate to $\approx 0.22\%$, and the corrections-pass false-positive rate to $\approx 0.3\%$.

Two cross-checks are worth flagging. First, the *student-perceived* per-rubric-item first-pass false-positive rate ($\approx 0.6\%$) is roughly an order of magnitude larger than the *instructor-confirmed* per-rubric-item rate from the regrade audit (0.04–0.08%, §4.1). Much of this gap is explained by students who noticed a deduction, disagreed with it, but did not

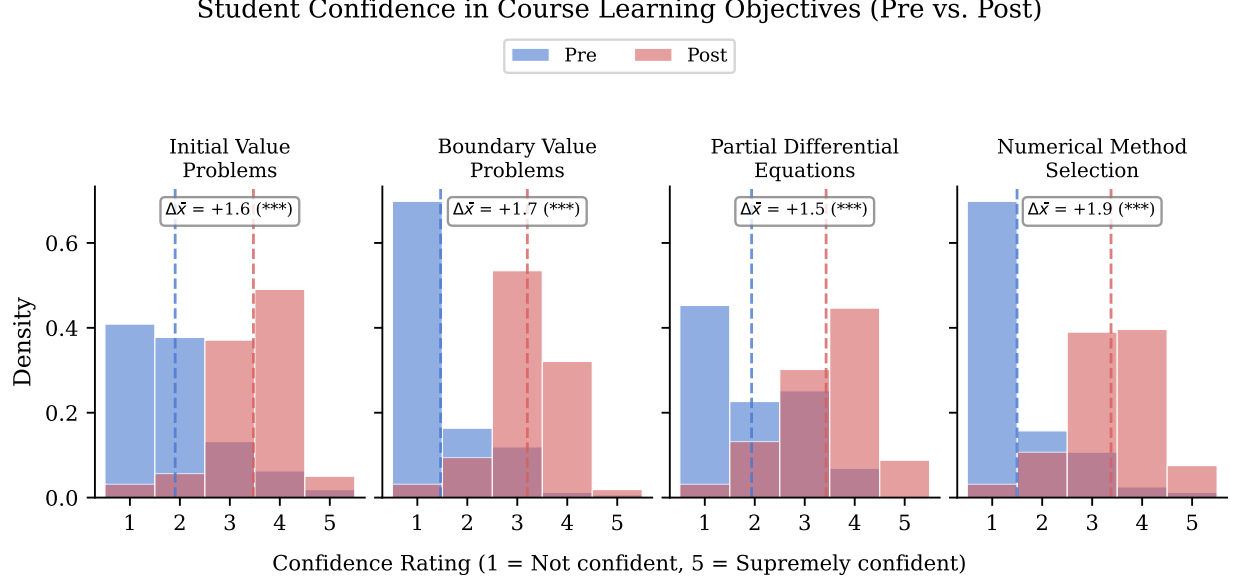


Figure 1: Retrospective pre/post confidence distributions on the four ME 373 learning objectives (1 = not confident at all, 5 = supremely confident). $N = 159$; dashed vertical lines mark group means; all comparisons $p < 10^{-27}$ by Mann–Whitney U .

formally request a regrade; either because the point total did not affect their letter grade, or because the corrections-pass workflow already restored the credit. Both numbers are below the noise floor of human-TA grading consistency reported in the broader assessment literature, and we read them as upper and lower bounds on the true first-pass error rate rather than treating either as ground truth. Second, first-pass false positives outnumber corrections-pass false positives by roughly 2:1, consistent with the rubric-evolution dynamic described in §??: the second pass runs against a more mature rubric and catches fewer spurious deductions.

Student-Reported Grading Errors Across Entire Term

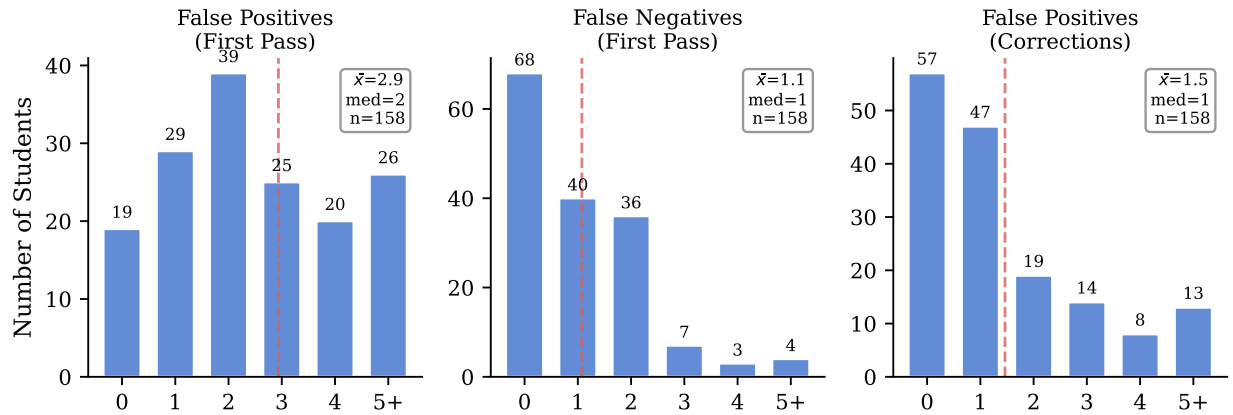


Figure 2: Student-reported LaTA grading errors over the full term. Left: first-pass false positives (incorrect deductions). Middle: false negatives (missed deductions). Right: corrections-pass false positives. Dashed line is the mean. Per-rubric-item rates derived from these counts are reported in the body.

LaTA perception Likerts. Figure 3 reports the three LaTA-perception Likert items as stacked percentages. Student ratings are honest and mixed. The LaTeX submission workflow itself was the most positively received component, with a mean of $M = 3.64$ ($SD = 1.13$) and 59.7% of respondents selecting 4 or 5; the feedback content ($M = 3.26$, $SD = 0.98$; 42.1% positive) and overall sentiment toward LaTA ($M = 3.16$, $SD = 1.09$; 40.3% positive) both sat slightly above the neutral midpoint but below the LaTeX-workflow item. In other words, students on average judged the transformation to a LaTeX-native workflow as more valuable than the autograder at the centre of that workflow; a finding we return to in §5.

LaTeX ramp-up cost. The paired time items (extra minutes per assignment spent on LaTeX) show a substantial reduction across the term (Figure 4). At the beginning of the term the cohort reported a mean extra time of $M = 87$ min per assignment ($SD = 63$, median = 60); by the end of the term this had fallen to $M = 47$ min ($SD = 48$, median = 30), a mean reduction of $\approx 46\%$. Median reductions are larger (50%) and better reflect the typical student, since the begin-of-term distribution carries a long right tail of students reporting ≥ 2 hours of extra LaTeX time on the first assignment. This pattern (a real up-front time cost that decays with practice) is consistent with the LaTeX-time theme in the open-ended essays (§4.3).

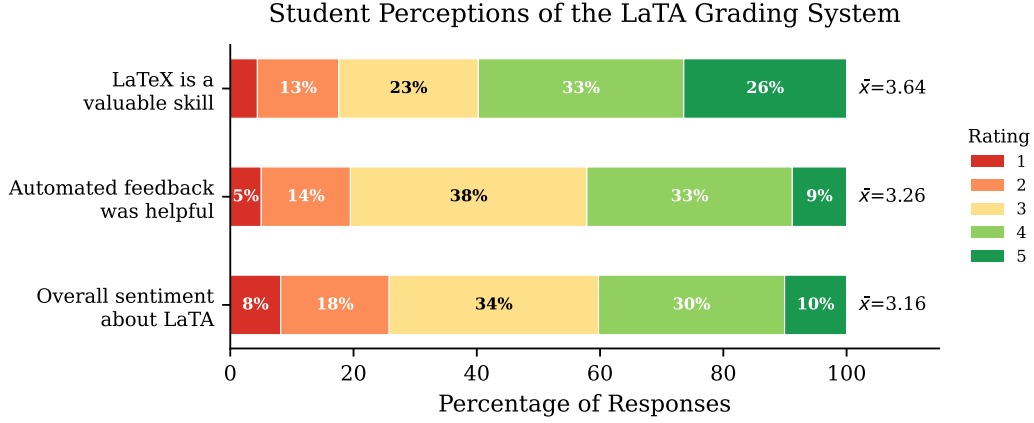


Figure 3: Student perception of three LaTA-related items (stacked diverging Likert, 1–5). The LaTeX workflow item draws more positive response than feedback quality or overall sentiment toward the system.

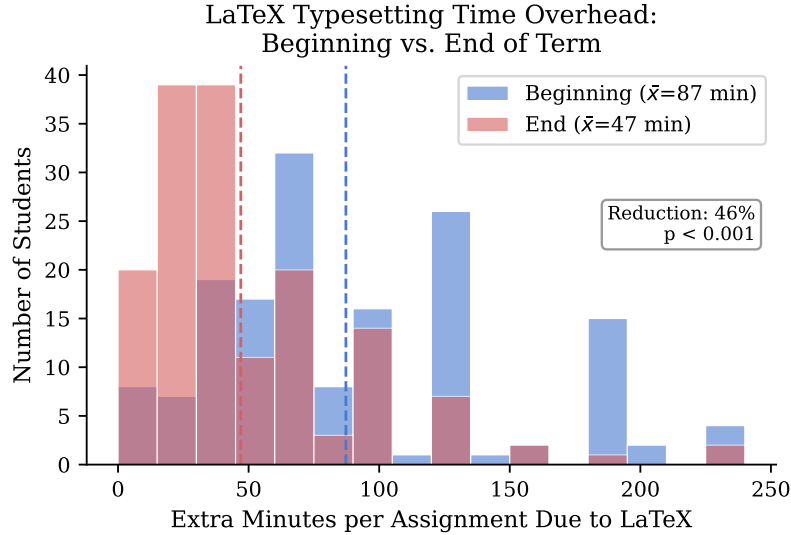


Figure 4: Student-reported extra time (in minutes per assignment) spent writing solutions in $\text{\LaTeX}$. Blue: beginning of term. Red: end of term. Medians halve; means drop by $\approx 46\%$.

Summary. Per-objective means, standard deviations, deltas, effect sizes, and p -values for the four learning objectives are collected in Table 1. The full numeric tabulation is reproducible from `analyze_feedback.py`.

Table 1: Pre/post confidence ratings across the four ME 373 learning objectives ($N = 159$). Mann–Whitney U test; effect size r_{rb} is the rank-biserial correlation.

Learning objective	Pre M (SD)	Post M (SD)	$\Delta\bar{x}$	r_{rb}	p
Initial value problems	1.91 (0.98)	3.47 (0.81)	+1.57	0.744	2.9×10^{-32}
Boundary value problems	1.47 (0.80)	3.20 (0.76)	+1.74	0.825	1.3×10^{-40}
Partial differential eqns.	1.94 (0.99)	3.43 (0.93)	+1.49	0.688	8.0×10^{-28}
Numerical method selection	1.50 (0.88)	3.38 (0.89)	+1.88	0.819	2.0×10^{-39}

4.3 Thematic coding of open-ended responses

Of the $N = 159$ respondents, 129 contributed a codable answer to the positive prompt and 131 to the negative prompt. Most of the dominant themes in both buckets are about course design broadly rather than LaTA specifically, and we report only the LaTA-relevant signal in detail here.

For context, we report the largest themes as the count of respondents in the relevant bucket whose essay was coded with that theme (each essay could receive multiple codes, so the counts do not sum to the bucket size). On the positive side ($n_+ = 129$ respondents), the largest non-LaTA themes were *homework-as-learning* (56 respondents, 43%), *lecture quality* (46, 36%), and *video lectures* (29, 22%). On the negative side ($n_- = 131$ respondents), the largest non-LaTA themes were *exam difficulty* (46 respondents, 35%), *workload* (34, 26%), and *lack of worked examples* (30, 23%). Each of these non-LaTA negative themes was endorsed by more respondents than either of the two LaTA-specific complaint themes (*LaTeX time cost*: 22, 17% of n_- ; *autograder errors*: 18, 14% of n_-). The takeaway from the broad coding is that LaTA was not the principal complaint of the term on either side of the ledger; the dominant complaints are familiar course-design concerns that long predate any autograder.

Figure 5 isolates only those free-text codes that directly reference LaTA or the LaTeX workflow, collapsed across the two prompts. The most-discussed LaTA-adjacent themes are *corrections workflow* (34 positive mentions, no negative mentions), *LaTeX valuable* (21 positive), *autograder errors* (18 negative), *LaTeX time cost* (22 negative), and *fast feedback* (11 positive). The smallest identifiable LaTA-related theme is *anti-AI principle* (4 negative respondents, 3% of n_-): students who objected to an AI system grading their work as a matter of principle, independent of whether the grades were correct. We consider this a real and persistent minority signal rather than noise, and return to it in §5. Across both prompts, many students wrote balanced, mixed-sentiment essays even in the “what worked well” box (45 positive-prompt essays coded as *mixed*, 67 negative-prompt essays coded as *mixed*), consistent with the mid-range overall-sentiment Likert mean of 3.16 reported above.

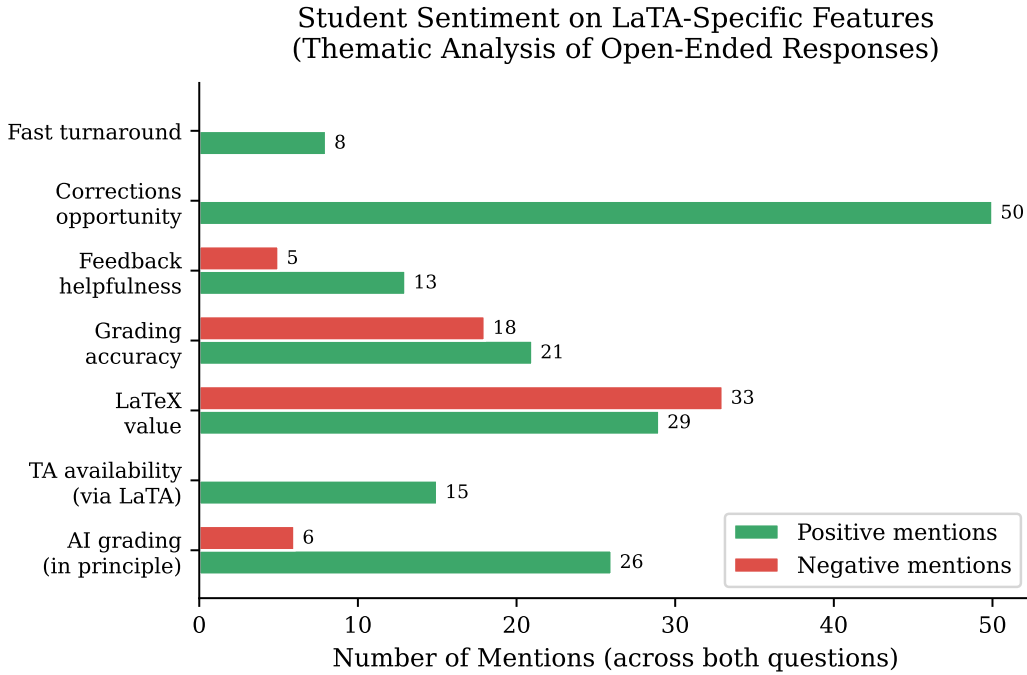


Figure 5: LaTA-specific free-text themes, stacked by sentiment (positive on top, negative on bottom). The corrections workflow is the single most-praised LaTA-adjacent feature and has no corresponding negative mentions; the anti-AI-principle category at $n = 4$ is the smallest identifiable LaTA-related theme.

Office-hours signal. A pair of Likert items asked students how helpful and how available TA and instructor office hours were during the term (Figure 6). Both items were rated positively ($M > 3.5$ on the 1–5 scale), consistent with the tripling of TA office-hour coverage described in §3.4. We flag this item here rather than in the LaTA-perception subsection because office hours are not part of LaTA; they are the bundled structural change that the TA-time savings enabled.

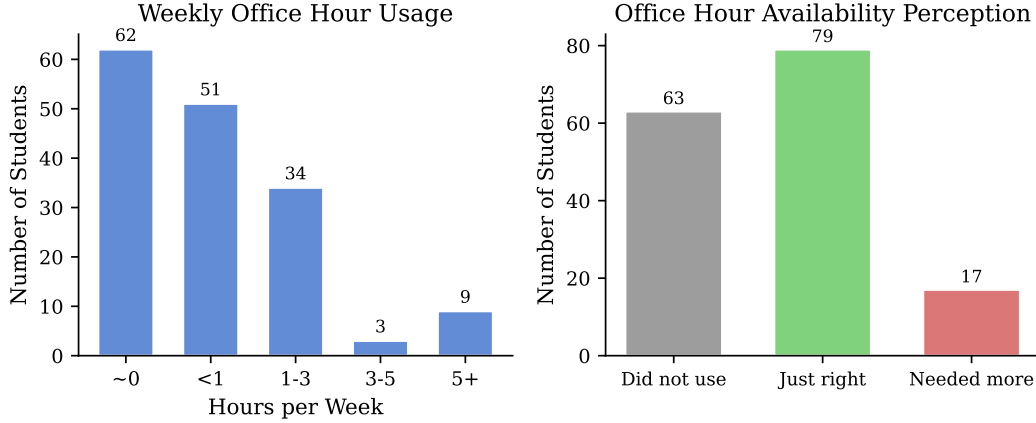


Figure 6: Student perception of office-hours helpfulness and availability (1–5 Likert). Both items receive $M > 3.5$, reflecting the $3\times$ expansion of TA coverage.

4.4 Between-cohort exam comparison

Mean exam scores for the LaTA-graded Winter 2026 cohort (enrollment 200; $n = 182$ sat the final) were higher than for the traditionally-graded Winter 2025 cohort (enrollment 181; $n = 157$ sat the final) by approximately 11% on the midterm and approximately 8% on the final, on a 0–100 scale. Because approximately one-third of exam items were updated between years and the new items were, in the instructor’s qualitative judgement, slightly harder, both deltas should be read as conservative lower bounds. We deliberately do not report inferential p -values on this comparison (§3.4); the point of the comparison is directional and coarse.

The between-cohort delta is consistent with, but stronger than, the within-cohort confidence gains reported in §4.2: exams are a harder criterion than retrospective self-report, and seeing a positive exam delta on top of a positive confidence delta mitigates the response-shift concern attached to the retrospective pre-test design. Two cautions apply. First, the Winter 2026 cohort received not only LaTA autograding but also a corrections workflow and approximately $3\times$ the TA office-hour coverage of the Winter 2025 cohort; the exam delta is attributable to the composite intervention, not to LaTA in isolation (§5). Second, a single-year, single-instructor comparison is not a controlled trial, and the delta reported here should not be read as a causal estimate of the autograder’s isolated contribution. What the exam delta does do is rule out the stronger negative hypothesis that moving to a LaTeX-native, LLM-graded workflow damaged student performance: within the precision of a one-year cohort comparison at this enrolment, it did the opposite.

5 Discussion

Read across the three evidence streams, Winter 2026 looks like a deployment that worked. LaTA processed ≈ 200 students for each of eight homework assignments on a single workstation, at 1–3 minutes of wall-clock time per submission and no end-to-end failures; the instructor-facing regrade rate held near 2.5% per assignment with half of requests judged valid; student confidence on every stated learning objective rose by more than one full Likert point; and a cohort whose exam items were, if anything, slightly harder than its predecessor’s outperformed that predecessor by roughly 11% and 8% on the midterm and final. We did not find evidence that replacing TA first-pass grading with an on-premises LLM pipeline harmed student learning. The more interesting question is what the evidence says about *how* it worked, and about the honest-mixed signal in the student-perception data.

5.1 The workflow, not the autograder, is what students valued

The single most striking pattern in the survey is that students rated the LaTeX submission workflow ($M = 3.64$, 59.7% positive) more highly than the quality of LaTA’s feedback ($M = 3.26$) or their overall sentiment toward LaTA as a grader ($M = 3.16$), and the thematic coding reproduced the same ordering: the corrections workflow was the most-mentioned positive LaTA-adjacent theme and attracted *zero* negative mentions, while LaTeX-valuable appeared 21 times positively and autograder-errors appeared 18 times negatively. In other words, students appear to be distinguishing the *framework* LaTA makes possible (typed submissions, fast turnaround, cheap corrections passes, feedback that lives in a document rather than a margin) from the *autograder at the core* of that framework.

This is, on reflection, what one should expect. The autograder itself is a commodity: any reasonable LLM pipeline that ingests LaTeX, applies a rubric, and returns structured JSON would produce feedback of roughly comparable quality given comparable rubrics. What LaTA does that most cloud-graders cannot is stay on-premises and release enough instructor time to fund the structural changes (corrections, tripled office hours) that students directly experience. The lesson for instructors considering an LLM-autograder deployment is that the autograder is probably not the headline feature; the headline feature is the budget that the autograder releases, provided the instructor uses that budget to change the course rather than simply to save time.

5.2 The student-perception gap: mixed is the honest reading

The three LaTA-specific Likert items all cluster slightly above the neutral midpoint, not solidly in “positive” territory. We take this mid-range signal at face value. Students who were charged an incorrect deduction — even one — and then had that deduction restored by a corrections pass or a regrade request are understandably not uniformly enthusiastic about the system that made the mistake, even if they acknowledge that the end-of-term point total was correct. Mean 3.16 overall sentiment is consistent with a cohort that experienced both the wins (fast turnaround, cheap corrections, consistent rubric application) and the costs (occasional misread notation, occasional LaTeX friction) of the deployment, and answered honestly. Treating these numbers as a failure of student buy-in would be a misread: the same respondents produced the large confidence gains in Block 1 and the thematic praise for the corrections workflow in Block 4.

The gap between student-perceived first-pass false positives ($M = 2.94$ per student over the term) and instructor-confirmed valid regrades from the audit (≈ 0.4 per student over the term) deserves a direct treatment. Part of the gap is definitional: a student’s self-reported false positive is any deduction they thought was wrong, whereas the audit counts only deductions that survived instructor review. Part of the gap is procedural: many students who disagreed with a first-pass deduction found the disagreement resolved by the corrections pass, and had no incentive to file a formal regrade. And part of the gap is almost certainly perceptual: the system was imperfect in ways that are noticeable to the student whose work was misread, and the audit undercounts those misreads. The useful posture is to read the two numbers as upper and lower bounds on the true first-pass error rate and to report both, rather than to claim the audit is the “real” number and the self-report is inflated.

5.3 The anti-AI-principle minority

A small but persistent minority of respondents (4 coded mentions, 3% of negative essays) objected to the use of AI to grade their work as a matter of principle, independent of whether the grades themselves were correct. We do not read this signal as a failure of the system but as an honest expression of a value judgement that the system cannot address by being more accurate. Any deployment of an LLM in a student-facing role will produce this minority, and the appropriate response is not to design around it but to acknowledge it. LaTA’s mitigations at the design level (full disclosure of AI grading in the syllabus, human-reviewable audit trails, a first-class regrade pipeline, on-premises data residency) are necessary but not sufficient: students who hold a principled objection to AI grading retain that objection even when all of the usual technical concerns (privacy, accuracy, recourse) have been addressed.

5.4 Attributing the exam-delta to the composite intervention

The between-cohort exam delta cannot be cleanly attributed to LaTA in isolation, because LaTA, the corrections workflow, and the tripling of TA office hours were introduced as a single package (§3.4). The most defensible reading of the delta is that the composite intervention improved outcomes by 8–11 percentage points relative to the previous year’s traditional deployment at the same institution with the same instructor, and that the pieces of the composite are causally interlocked: the autograder released the TA time that funded the extra office hours, and the typed-submission workflow is what made cheap corrections passes possible at all. A pure-LaTA counterfactual (LLM grading with no corrections workflow and no extra office hours) would probably still save instructor time, but the exam signal we report is about the full package. The cleanest way to separate the three components is a multi-section or multi-course replication in which they are introduced independently; we flag this as a high-priority future design (§7).

5.5 Generalisation envelope

The specific course we deployed to; an undergraduate numerical-methods course with heavy symbolic derivation, LaTeX submissions already permissible, and a 200-student enrollment, sits near the sweet spot for an LLM autograder: problems are long enough that copying a reference answer is not a tight rubric, short enough to fit in a 64k-token context, and structured enough that a binary-scored rubric reliably captures the decisions a human TA would make. Two deformations of the setting would make the approach straightforwardly transferable and one would make it harder.

Straightforwardly transferable: *adjacent symbolic-derivation courses* (differential equations, continuum mechanics, introductory fluids, heat transfer, controls) where solutions are worked algebraically and an answer is a short expression or a sentence of physical interpretation. The dominant effort in such a deployment is rubric authoring, not pipeline engineering; the pipeline described here runs as-is. Also transferable: *LaTeX-friendly disciplines beyond engineering* (physics, applied mathematics, economics with a formal component), provided the instructor is willing to require LaTeX submissions.

Harder: courses whose core deliverable is a *plot, a piece of running code, or a schematic*. LaTA grades plots through the `points_awarded_elsewhere` hand-graded path (§??); it does not run student code or inspect a figure. Tool-calling and multimodal LLMs are the obvious future extensions, and both are explicit work items for LaTA. A separate concern is *natural-language reasoning problems* (e.g. a design justification essay), where binary rubric items are a worse fit than they are for a derivation and the value of an LLM grader depends more sensitively on the rubric-author’s skill. These are not unreachable settings, but they are not the setting in which LaTA was validated.

6 Limitations

The evidence reported in this paper has six limitations worth naming explicitly.

Single-instructor, single-course, single-year deployment. All data come from one section of ME 373 taught by the author at Oregon State University during Winter 2026, compared against the corresponding section taught by the same author during Winter 2025. The strongest threat to generalisation is simply that one instructor’s tastes in rubric authoring, feedback style, and class culture are bundled into every measurement we report. A deployment by a different instructor (even of the same tool against the same syllabus) would produce different perception scores, a different regrade-request distribution, and probably a different ceiling on what the autograder can score reliably. The results reported here should be read as an existence proof that the tool can carry a full-replacement deployment at this scale with these outcomes, not as an estimate of the effect size an arbitrary instructor should expect.

Composite intervention. As detailed in §3.4 and §5.4, the exam-score delta against the Winter 2025 cohort is attributable to the full package of LaTA autograding *plus* the corrections workflow *plus* a tripling of TA office hours, and a single year of post-hoc data cannot separate the three. The delta we report is the composite effect and we read it as such throughout. A multi-section or multi-year replication that introduces the three components independently is the clean way to disentangle them.

Retrospective pre-test. The pre/post confidence gains in Block 1 were collected through a single-administration retrospective pre-test. This design eliminates the response-shift bias that afflicts separate first-week and last-week surveys [27] but introduces its own well-documented tendency to inflate apparent gains: students who finish a course with greater confidence retrospectively downrate their pre-course confidence compared to what they would have self-reported at the beginning. Every gain we report exceeds one full Likert point, and the between-cohort exam comparison moves in the same direction, so the qualitative conclusion is unlikely to be an artefact of the instrument; nonetheless the precise gain magnitudes should not be read as clean pre/post differences.

Unpaired analysis of paired pre/post items. Because the survey was anonymous and delivered in a single administration, we cannot match a respondent’s pre-term confidence rating to the same respondent’s post-term rating, and we analysed the two distributions with the Mann–Whitney U test as if they were independent samples (§3.5). A paired test (Wilcoxon signed-rank) would use the actual dependence structure and be slightly more powerful; conditional on the anonymous design, the reported p -values are a small and conservative approximation. This does not change any qualitative conclusion at the magnitudes reported, but future survey instruments should include a non-identifying session token that permits paired analysis.

Single-coder thematic analysis. The open-ended essay responses were coded by one rater (the instructor–author) against a codebook developed on a random 20% sample (§3.3). No inter-rater-reliability statistic is available, and the coder is not blind to the intervention. The codebook and the full response-level coding are available upon request so that interested readers can apply their own coding scheme; readers who would prefer a second rater can treat the thematic counts here as a descriptive signal to be checked against the Likert items, not as a standalone claim.

No head-to-head comparison with alternative graders. This study reports a deployment of one specific autograder configuration (gpt-oss:120b on Ollama at a 65k-token context, with a binary-scored rubric and the prompt-injection defences in §2.1). We do not compare LaTA’s grading decisions against those of a cloud-hosted LLM, against a different

local model, against a human TA on the same submissions, or against LaTA with the injection defences disabled. Each of these is a natural follow-up study, and each would answer a question this deployment-focused paper does not. In particular, the claim that the pipeline is robust to prompt injection is a claim-by-construction (the five layers are present in the source code and the audit log records no successful injections during the term) rather than a red-team validation.

Plot hand-grading carve-out. LaTA did not grade the plot-only subproblems of the assignments; those were hand-graded visually by the instructor using the `points_awarded_elsewhere` mechanism (§??). The accuracy numbers we report cover the LLM-scored components only. Future versions that use tool-calling to run student code or multimodal LLMs to inspect figures would close this gap, but the current deployment does not speak to them.

7 Conclusion

LaTA is an on-premises, FERPA-compliant, full-replacement autograder for handwritten-style STEM coursework. It ingests LaTeX submissions from Gradescope, segments them, grades them against an instructor-authored YAML rubric using a large local reasoning LLM on a single workstation, and emits per-student PDF feedback that self-heals through LaTeX compilation retries. The pipeline is deterministic enough in its structural decisions (typed schemas, binary-scored rubrics, regex-first segmentation) and defensive enough in its LLM-facing surface (five-layer injection defence, LLM-fixer for compilation errors, audited regrade trail) to run unattended on 200-student assignments for a full term.

Deployed through one section of a numerical-methods course at Oregon State University during Winter 2026, LaTA graded every submission of eight homework assignments with no end-to-end failures, an instructor-facing valid-regrade rate of approximately 2.5% per assignment, and 1–3 minutes of wall-clock time per submission. Relative to the same instructor’s previous traditionally-graded cohort, the LaTA-graded cohort reported large confidence gains on all four stated learning objectives and outperformed the previous cohort by 11% on the midterm and 8% on the final on a shared exam structure, with the exam delta biased conservatively by the slightly harder new items. Student perception of the autograder itself was mid-range and honest; perception of the LaTeX submission workflow and the corrections pass it made possible was substantially more positive. The most-cited positive theme in open-ended responses was the corrections workflow, which drew zero negative mentions.

We read the evidence as saying that the autograder is useful primarily because it releases instructor and TA time, and that instructors considering a deployment of this kind should plan the deployment around what to do with that released time (in our case, corrections passes and expanded office hours) rather than around the autograder in isolation. The next priorities for LaTA itself are tool-calling to run student code (closing the plot-grading carve-out), multimodal grading for figures, and a paired-design survey instrument that supports a within-subject pre/post analysis. The clearest external priority is a multi-instructor, multi-course replication that disentangles LaTA, the corrections workflow, and the expanded office hours as independent interventions, so that future deployments can be costed and planned component by component rather than as a single bundle.

The LaTA source code, the survey instrument, the analysis script, and the thematic coding are available as described in the Data and code availability statement below.

CRedit authorship contribution statement

Jesse A. Rodríguez: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing — original draft, Writing — review & editing, Visualization, Supervision, Project administration.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Anthropic Claude (Sonnet and Opus, including the agentic Claude Code system for original code development) for draft editing and code documentation review. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data and code availability

The LaTA source code is released under AGPLv3 at <https://github.com/JesseRodriguez/LaTA>. Student-level grading data are FERPA-protected and cannot be shared; aggregate, de-identified statistics reported in §4 as well as other instruments are available from the author upon reasonable request.

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